Abstract

The aim of this sub-project is to develop an image processing component to support low-cost automated greenhouses. The image processing application must enable the greenhouse system to establish various aspects about the plants from the images taken. These aspects include plant health, plant growth and plant disease identification from the images of different parts of the plants in the greenhouse.

Fill out the introduction.

What is the purpose of this section?

- What is plant phenotyping?
- Discuss what can be measured by plant phenotyping?
- Why bother about plant phenotyping?

Context for plant phenotyping? It is of vital importance in plant breeding to be able to propogate desirable heritable traits from plants through controlled human action [9]. This is especially important in the context of food security where a cultivars resistant to adverse abiotic and biotic stresses affect the final overall crop yield. The literature indicates that plant phenotyping has become an effective tool in plant breeding methods. [1].

What is plant phneotyping Plant phenotyping is then the evaulation of phenotypes in plants with the main goal of producing quantative data to study the dynamic interaction between a plant and its environment [3].

Definitons 2
The phenotype can be seen as the combination of all the morphological, physiological, anatomical, chemical, developmental, and behavioral characteristics that, when put together, represent the individual organism. Phenotyping, which is the process of characterizing the phenotype [5]

Definitons 3
For the purposes of this review, we refer to phenotyping as the set of methodologies and protocols used to measure plant growth, architecture, and composition with a certain accuracy and precision at different scales of organization, from organs to canopies. The goal of modern plant phenotyping is to deliver quantitative data on the dynamic responses of plants to the environment. [3]

Definitons 4
The broad evaluation of complex plant traits like growth, development, tolerance, resistance, architecture, physiology, ecology, and yield as well as the fundamental measurement of individual quantitative parameters that serve as the foundation for more complex traits is known as plant phenotyping [8]

Definitons 5
Plant phenotyping is the comprehensive assessment of complex plant traits such as growth, development, tolerance, resistance, architecture, physiology, ecology, yield, and the basic measurement of individual quantitative parameters that form the basis for more complex traits [7]

Explanation of relevance of plant phenotyping for current study? The term phenotype was first employed by Johannsen [6]. Who stated that a phenotype is all types of organism which are idenfitied from one antoher by inspection or by some measuremment. So it is an obserable primitive trait or a complex trait of a plant that can be identified visually or measured and evaulated numerically. Therfore the data produced by plant phenotyping assist in choosing the genetic material (germplasm) that will enable plant breeders to induce traits in cultivars which are optimal for various diffrent growing environments (like water scarce environments).

Why plant phenotyping in the context of this study? Conventional plant plantphenotyping suffers from scalabality issues and is often constrained by costs, time, and human effort required. Furthermore many methods are destructive to the cultivar, making it hard to replicate, and compare measurements of phenotypes over time through the ontogensis of a cultivar. However in the past decade image processing techniques have been applied successfully to aquire phenotypical data in a non-destructive matter [2].. There exists many different image techniques that have been applied in greenhouses and in the field which allow for the assesment of resistance to disease, biomass, soli salinity, etc using various differenr sensors, and techniques.

What affects plant growth?
Plant biomass is affected by stresses and disturbances [4]. A stressor is anything which stymies the growth of plant biomass. Examples of stressors include things like water scarcity, inadequate amounts of light, and unfavourable temperatures. A disturbance is anything which causes some level of destruction to the plant. Examples of disturbances include pathogens, fires, frost, and pests.

How can we measure plant stresses and disturbances?
Stresses and disturbances of plants are induced by environmental conditions the organisms are placed in, and therefore transitively induce phenotypical traits. We can exploit this fact by measuring plant phenotypical traits, we can infer the prescence of potential stresses and disturbances on cultivars. For example thermal imaging allows to meausre the phenotype of plant temperature or plant transpiration, which in turn can give us an indication of soil salinity. We indirectly infer environmental conditions based on phenotypical traits

What physical traits of a plant can be measured phenotypically?

Conceptual basis Typically in the context of plant breeding we care about plant yield, resistant to abiotic and biotic stresses. That is in general we seek plants which are fit. In the literature of the term trait is used to describe individual scoped properties, communit scoped properties end even environmental properties. trait is any morphological, physiological or phenological feature measurable at the individual level, from the cell to the whole-organism level, without reference to the environment [10]. Plant phenotyping reflects this in its achievements and developments over the past decade. The key is environmental related performance metrics are derived from individual plant traits. So these phenotypical traits act as proxies that indirectly preditc plant fitness and performance.

Morphological traits About plant sturcture, which affects its performance and gives indication regarding its health

- Leaf width
- Number of leaves
- Leaf inclination angle
- Leaf area
- Root system architecture (e.g root diameter)
- Plant shoots
- Fruit
- Flowers

Physiological traits About how plant operates in repsonse to its growing conditions

- Stomotal conductance
- Chlorophyll content of leaf
- Plant water status
- Plant photosynthesis (metablism)
- Nutrient content

Phenological traits These are phenological traits (when certain biological events occur that depend on climates)

- When fruit or flowers are yielded
- Yield prediction

Phenotypical traits as proxy measurements for plant stress measurement How can these traits be used to predict abiotic and biotic stresses

What is the purpose of this section?

- Discuss image processing in the context of plant phenotyping
- Discuss image acquistion techniques
- Discuss image processing techniques used in plant phenotyping

How is image processing applied in plant phenotyping?
Fill out this section

What can be used for image acquistion?

- Thermal imaging sensors
- Hyperspectral sensors
- RGB camera
- Tomography
- Flurescence sensors

Fill out the applications in low cost greenhousing techniques secton.

Fill out the conclusion section.

References

[1] Achala Anand, Madhumitha Subramanian, and Debasish Kar. "Breeding Techniques to Dispense Higher Genetic Gains". In: *Frontiers in Plant Science* 13 (Jan. 19, 2023), pp. 4–5. ISSN: 1664-462X. DOI: 10.3389/fpls.2022.1076094. URL: https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2022.1076094/full (visited on 05/16/2025).

[2] Jayme Garcia Arnal Barbedo. "Digital Image Processing Techniques for Detecting, Quantifying and Classifying Plant Diseases". In: *SpringerPlus* 2.1 (Dec. 7, 2013), p. 660. ISSN: 2193-1801. DOI: 10.1186/2193-1801-2-660. URL: https://doi.org/10.1186/2193-1801-2-660 (visited on 05/11/2025).

[3] Fabio Fiorani and Ulrich Schurr. "Future Scenarios for Plant Phenotyping". In: *Annual Review of Plant Biology* 64.1 (Apr. 29, 2013), pp. 267–291. ISSN: 1543-5008, 1545-2123. DOI: 10.1146/annurev-arplant-050312-120137. URL: https://www.annualreviews.org/doi/10.1146/annurev-arplant-050312-120137 (visited on 05/14/2025).

[4] J.P Grime. "Evidence for the Existence of Three Primary Strategies in Plants and Its Relevance to Ecological and Evolutionary Theory". In: *The American Naturalist* 111.982 (1997), p. 1169. DOI: 10.1086/283244. URL: https://www.journals.uchicago.edu/doi/abs/10.1086/283244 (visited on 05/19/2025).

[5] Qiaosheng Guo and Zaibiao Zhu. "Phenotyping of Plants". In: *Encyclopedia of Analytical Chemistry*. John Wiley & Sons, Ltd, 2014, pp. 1–15. ISBN: 978-0-470-02731-8. DOI: 10.1002/9780470027318.a9934. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/9780470027318.a9934 (visited on 05/14/2025).

[6] W. Johannsen. "The Genotype Conception of Heredity". In: *The American Naturalist* 45.531 (Mar. 1911), pp. 129–159. ISSN: 0003-0147, 1537-5323. DOI: 10.1086/279202. URL: https://www.journals.uchicago.edu/doi/10.1086/279202 (visited on 05/14/2025).

[7] Lei Li, Qin Zhang, and Danfeng Huang. "A Review of Imaging Techniques for Plant Phenotyping". In: *Sensors (Basel, Switzerland)* 14.11 (Oct. 24, 2014), pp. 20078–20111. ISSN: 1424-8220. DOI: 10.3390/s141120078. PMID: 25347588. URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4279472/ (visited on 04/20/2025).

[8] Semira Ebrahim Mohammed. "Review on Plant Phenotyping". In: (). URL: https://www.abrinternationaljournal.org/abstract/review-on-plant-phenotyping-1102322.html (visited on 05/05/2025).

[9] Thomas J. Orton. "Introduction". In: *Horticultural Plant Breeding*. Elsevier, 2020, pp. 3–7. ISBN: 978-0-12-815396-3. DOI: 10.1016/B978-0-12-815396-3.09986-8. URL: https://linkinghub.elsevier.com/retrieve/pii/B9780128153963099868 (visited on 05/16/2025).

[10] Cyrille Violle et al. "Let the Concept of Trait Be Functional!" In: *Oikos* 116.5 (2007), pp. 882–892. ISSN: 1600-0706. DOI: 10.1111/j.0030-1299.2007.15559.x. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/j.0030-1299.2007.15559.x (visited on 05/20/2025).